POLARIS BRIDGE ENGINEERING LLC

CIVIL / STRUCTURAL & BRIDGE ENGINEERING

Bridge Design • Rehabilitation • Construction Engineering • Inspection • Load Rating • Building Structures

Peng Chen, P.E. | Founder & Principal | (732) 491-7038 | pengchen@polarisbridgeengineering.com | www.polarisbridgeengineering.com

CORE CIVIL / STRUCTURAL SERVICES

- Bridge structural design: steel, reinforced concrete and prestressed concrete structures
- Bridge rehabilitation, repair design and retrofit detailing
- Bridge construction engineering, erection support and construction-stage analysis
- Temporary works: shoring, support towers, sheet piles and temporary retaining structures
- Bridge inspection, NBIS condition assessment and deficiency documentation
- Bridge load rating and structural capacity evaluation using LRFR / LFD / LRFD methods
- Commercial building structural design, evaluation and retrofit support
- Structural condition assessment for buildings, parking structures and facilities
- Foundations, piles, retaining walls, culverts and rigid-frame structures
- Shop drawing review, RFIs, field-design support, constructability and QA/QC

CREDENTIALS

- Professional Engineer: VA, MD, DC, NY and NJ
- FHWA NBIS Certified Bridge Inspector
- FHWA NBIS Certified Tunnel Inspector
- M.S., Civil Engineering - Rutgers University
- 20+ years of civil/structural, bridge and infrastructure experience

TECHNICAL RESOURCES

OpenBridge Designer / Structural WorkSuite | AASHTOWare BrR | MIDAS | CSI Bridge | SAP2000 | LARSA | STAAD | LEAP Bridge | AutoCAD | MicroStation | Mathcad

CODES & STANDARDS

AASHTO | AREMA | FHWA / NBIS | ACI 318 | AISC | ASCE 7 | IBC | State DOT criteria and specifications

PROCUREMENT PROFILE

Legal Name: Polaris Bridge Engineering LLC
Firm Type: Virginia limited liability company
Status: Virginia-Certified DBE / MBE, MWAA Certified SLBE
NAICS: 541330 - Engineering Services

COMPANY PROFILE

Polaris Bridge Engineering LLC provides principal-led civil/structural and bridge engineering support to prime consultants, contractors, public owners, property owners and facility managers. The firm focuses on practical engineering solutions for bridge design, rehabilitation, construction-phase support, inspection, load rating, commercial building structures and structural condition assessment.

BRIDGE DESIGN & REHABILITATION CAPABILITY

Peng Chen, P.E. brings extensive bridge design and rehabilitation experience across VDOT, NYSDOT, NJDOT, ODOT, TxDOT, TDOT, WMATA, CSX and other transportation programs. Representative work includes steel plate girders, curved steel girders, prestressed concrete girders, abutments, piers, pile foundations, retaining walls, culverts, deck replacement and structural repair design.

- VDOT statewide bridge assignments included Route 189, Route 671, Route 641 and Route 620 bridge/culvert replacement work.
- WMATA rehabilitation work included bearing replacement, jacking and blocking, steel-section retrofit, joint replacement, painting, concrete patching and crack repair.

BRIDGE CONSTRUCTION ENGINEERING

Polaris offers contractor-oriented structural engineering support for bridge construction, temporary works, erection planning, construction-stage analysis, shop drawing review, RFIs and field-design issues.

- NYSDOT I-287 Cross Westchester Expressway Interchange 8: prior construction engineering support to ECCO III included steel-girder erection, precast box/rigid-frame culvert erection, temporary support towers, sheet piles, jacking pits, slope-stability checks and temporary retaining-wall design.
- TDOT I-440 Design-Build: construction-stage analysis of a curved steel box-girder bridge during widening and erection.

BRIDGE INSPECTION & LOAD RATING

Experience includes NBIS bridge inspection, structural condition assessment, deficiency documentation, QA/QC of inspection reports and bridge load rating for steel, concrete, prestressed concrete and other transportation structures.

- Managed load rating and condition-assessment review for 162 WMATA aerial and pedestrian bridges.
- Performed or led inspection/report QA/QC for approximately 120 bridges and culverts in the City of Suffolk and nine bridges plus one culvert on the VDOT Dominion Boulevard Improvements project.

COMMERCIAL BUILDING & STRUCTURAL ASSESSMENT

Polaris provides civil/structural engineering for commercial and institutional buildings and facilities, including structural framing evaluation, equipment support design, condition investigations, repair concepts and retrofit support. Services can include assessment of deteriorated or distressed structural elements, documentation of deficiencies, development of repair recommendations and design of practical structural modifications.

REPRESENTATIVE STRUCTURAL / INFRASTRUCTURE EXPERIENCE

- I-64 Hampton Roads Bridge-Tunnel Expansion: trestle bridge, sign structure, rehabilitation/repair engineering and design QA/QC.
- NYSDOT Gowanus Expressway: emergency/red-flag repair design, member load rating, repair detailing and constructability evaluation for a major steel viaduct.
- CSX Bridge over 3rd Avenue: railroad bridge design including temporary jumper span, temporary shoring and pile foundations.
- MTA-LIRR East Side Access: lead structural engineering/team leadership for the 90% structural submission of the new three-level station beneath Grand Central Terminal.
- Port Authority of NY & NJ facilities: structural inspection experience involving airport garages, terminal buildings, high-mast light towers and marine-terminal structures.

WHY POLARIS

- Direct principal involvement from scoping through final deliverables
- Civil/structural capability spanning bridges, transportation infrastructure and commercial buildings
- Strong contractor-side construction engineering and temporary-works experience
- Practical recommendations grounded in design, inspection, load rating and field construction experience
- Flexible DBE subconsultant support for prime consultants and contractors

Available for teaming and subconsultant assignments: Bridge | Highway | Transit | Commercial Buildings | Facilities